

A counterexample to the prime-order-factor conjecture for unimodal cyclotomic generating functions

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Abstract

We exhibit an explicit unimodal basic cyclotomic generating function of degree 216 whose cyclotomic factors have only the composite orders 4, 9, 25 and 30. A complete positive centered q -integer expansion certifies strict unimodality. The example refutes the prime-order-factor conjecture of Billey and Swanson. The actual polynomial, coefficient certificate, basic quotient identity and universal prime exclusion have complete Lean proofs.

1 The conjecture and the counterexample

Billey and Swanson [1], Conjecture 48, propose that every nonconstant unimodal basic cyclotomic generating function has a prime-order cyclotomic factor. Here $[m]_q = 1 + q + \cdots + q^{m-1}$ for $m \geq 1$; a basic CGF is a polynomial with nonnegative integer coefficients that is a quotient of products of these q -integers, without an additional scalar or monomial factor. The conjecture imposes no squarefreeness condition or exclusion of powers. The predicted factor is Φ_p for a prime p , not an arbitrary prime-power order.

Theorem 1. *The polynomial*

$$F(q) = (\Phi_4(q)\Phi_9(q)\Phi_{25}(q)\Phi_{30}(q))^6$$

is a nonconstant basic CGF of degree 216. Its integer coefficients are strictly positive, strictly increasing up to degree 108 and strictly decreasing thereafter. No Φ_p with p prime divides it. Consequently Conjecture 48 of [1] is false.

2 Factorization and the basic-CGF identity

Put $B = \Phi_4\Phi_9\Phi_{25}\Phi_{30}$. The factors are

$$\Phi_4(q) = 1 + q^2,$$

$$\Phi_9(q) = 1 + q^3 + q^6,$$

$$\Phi_{25}(q) = 1 + q^5 + q^{10} + q^{15} + q^{20},$$

$$\Phi_{30}(q) = 1 + q - q^3 - q^4 - q^5 + q^7 + q^8.$$

The identity $q^m - 1 = \prod_{d|m} \Phi_d(q)$ implies

$$B(q) = \frac{[4]_q[9]_q[25]_q[30]_q}{[1]_q[6]_q[10]_q[15]_q}. \quad (1)$$

Indeed, after cancelling the four copies of $q - 1$ on each side, the cyclotomic factors of the numerator are those with orders dividing 4, 9, 25 or 30, with multiplicities. Cancelling the factors whose orders divide 1, 6, 10 or 15 leaves exactly 4, 9, 25 and 30 once each. The denominator is a nonzero polynomial. Raising (1) to the sixth power supplies the required quotient representation for F . The remaining requirement of coefficient nonnegativity is verified next.

3 A complete finite coefficient certificate

All four factors are reciprocal. Consequently if $F = \sum_{j=0}^{216} a_j q^j$, then $a_j = a_{216-j}$. Exact integer multiplication yields

$$F(q) = \sum_{j=0}^{108} d_j q^j [217 - 2j]_q. \quad (2)$$

The complete list of weights follows; each row starts at its displayed index and lists consecutive values. The last row has only four values.

j	Consecutive values $d_j, d_{j+1}, \dots$				
0	1	5	15	35	64
5	96	126	150	180	238
10	326	426	501	531	567
15	701	991	1371	1692	1866
20	1980	2326	3071	4023	4806
25	5124	5250	5856	7269	9147
30	10585	11189	11655	13079	15982
35	19290	21585	22779	24183	27587
40	32545	36825	39095	40213	43527
45	50235	57570	62100	62948	64882
50	72555	84635	95020	97758	96567
55	100479	113595	130975	140138	138138
60	135060	142464	162645	180663	184173
65	177591	176555	194149	218922	230692
70	223826	212586	221560	248612	269481
75	265241	241396	232836	254723	286675
80	295956	268346	238156	241074	275394
85	304410	288381	244691	218995	235689
90	273250	275390	232581	184211	174619
95	208125	227485	197435	138246	102744
100	124275	158475	150360	91356	28794
105	23500	59600	78150	38406	

Identity (2) is a finite polynomial certificate: expanding the four displayed factors and raising their product to power six gives its left-hand side, and expanding the 109 specified centered q -integers gives its right-hand side. The coefficient list and an independent integer-arithmetic verifier of both

sides accompany this paper; the complete identity is also proved in Lean. For $0 \leq k \leq 108$, a term with index j contributes to degree k exactly when $j \leq k$: its support is $[j, 216 - j]$. Hence

$$a_k = \sum_{j=0}^k d_j.$$

Every listed d_j is strictly positive, so the first-half coefficients are strictly positive and strictly increasing. Reciprocity gives the strictly decreasing second half. This proves the required nonnegativity and strict unimodality, with a unique peak at 108. The coefficient sum is $30^6 = 729000000$ and $a_{108} = 11434392$.

4 Excluding every prime-order factor

Cyclotomic polynomials are irreducible over $\mathbb{Q}$, and distinct cyclotomic polynomials are relatively prime. The complete factorization of F has only orders 4, 9, 25 and 30, none prime. Thus Φ_p does not divide F for any prime p . Equivalently, a primitive p th root cannot have any of these four distinct composite orders. This proves the universal prime exclusion without extrapolating from finitely many primes. Finally $\deg F = 6(2 + 6 + 20 + 8) = 216 > 0$, so $F \neq 1$. All assertions of the theorem follow.

5 Reproduction and scope

Run `python3 verify_counterexample.py` in this paper's directory. The standard-library checker reconstructs the cyclotomic factors from $q^m - 1$, verifies (1), multiplies out all 217 coefficients, checks all 109 weights and independently reconstructs (2). It also checks exact remainders for every possible prime up to 217; larger prime-order divisors are excluded by degree. That finite check supplements the universal factorization proof above. There is no numerical-root computation. No minimum-degree or worldwide-first claim is made, and the distinct later HSOP-membership problem is not the conjecture addressed here.

6 Formal verification and reproduction

The principal theorem has a complete Lean proof. The precise scope is: The explicit degree-216 polynomial, strict unimodality, basic q-integer quotient, and exclusion of every prime-order cyclotomic factor. No degree-minimality claim. The public proof source snapshot is fixed at commit `02af7db99db789fecb0d5f5f8f37b1a4e40762a0` in <https://github.com/mxym/math>. The proof packages, with their own dependency locks, audit records and reproduction instructions, are:

- `formalizations/cyclotomic-prime-factor-counterexample`.

The main formal entry points are:

- `CyclotomicCounterexample.explicit_counterexample`.
- `CyclotomicCounterexample.conjecture48_false`.
- `CyclotomicCounterexample.F_centered_certificate`.
- `CyclotomicCounterexample.F_qInteger_quotient_certificate`.

Lean is fixed to version 4.34.1, compiler commit 5045d0056413266e57c625dcd7c365b10e377c52; Mathlib is fixed to d13f23b723b8a846827a245b89c10fc7d3f11612. The audited mathematical roots depend only on the standard axioms `propext`, `Classical.choice`, and `Quot.sound`. The archived audits state their actual compilation and replay scopes; publication of this typeset version does not imply a new execution of every previous empty-kernel replay. The companion README maps each printed principal result to its formal theorem and explains how to repeat verification.

Research and preparation disclosures

This research, formalization and manuscript preparation used AI assistance. The author is responsible for the manuscript. The principal claims are supported by the readable proofs and the public Lean artifacts specified above. Formal verification is not external human peer review; no such review is claimed. No external funding was received. No worldwide priority or journal acceptance is asserted.

References

- [1] S. C. Billey and J. P. Swanson, *Cyclotomic Generating Functions*, Electronic Journal of Combinatorics **31**(4) (2024), P4.4. <https://doi.org/10.37236/12687>. Pinned manuscript: <https://arxiv.org/abs/2305.07620v3>, Definitions 1, 3 and 45; Conjecture 48, printed page 32.